FRONT 15.5 cm 25 cm

ELONZA-100

ERECTILE DYSFUNCTION DRUG

Each film coated tablet contains:
Sildenafil citrate eq. to
Sildenafil 100 mg

PRODUCT DESCRIPTION:
Royal blue, rhombic, biconvex, film coated tablet with engraved 100 on one side and scored on the other

MECHANISM OF ACTION:
Pharmacology
Sildenafil is an oral therapy for erectile dysfunction. In the natural setting, i.e. with sexual stimulation, it restores impaired erectile function by increasing blood flow to the penis.
The physiological mechanism responsible for erection of the penis involves the release of nitric oxide (NO) in the corpus cavernosum during sexual stimulation. Nitric oxide then activates the enzyme guanylate cyclase, which results in increased levels of cyclic guanosine monophosphate (cGMP), producing smooth muscle relaxation in the corpus cavernosum and allowing inflow of blood. Sildenafil is a potent and selective inhibitor of cGMP specific phosphodiesterase type 5 (PDE5) in the corpus cavernosum, where PDE5 is responsible for degradation of cGMP. Sildenafil has a peripheral site of action on erections. Sildenafil has no direct relaxant effect on isolated human corpus cavernosum but potently enhances the relaxant effect of NO on this tissue. When the NO/cGMP pathway is activated, as occurs with sexual stimulation, inhibition of PDE5 by Sildenafil results in increased corpus cavernosum levels of cGMP. Therefore sexual stimulation is required in order for Sildenafil to produce its intended beneficial pharmacological effects.
Studies *in vitro* have shown that Sildenafil is selective for PDE5, which is involved in the erection process. Its effect is more potent on PDE5 than on other known phosphodiesterases. There is a 10-fold selectivity over PDE6 which is involved in the phototransduction pathway in the retina. At maximum recommended doses, there is an 80-fold selectivity over PDE1, and over 700-fold over PDE 2, 3, 4, 7, 8, 9, 10 and 11. In particular, Sildenafil has greater than 4,000-fold selectivity for PDE5 over PDE3, the cAMP-specific phosphodiesterase isoform involved in the control of cardiac contractility.
Pharmacokinetics
Absorption/ Bioavailability: 30 to 120 minutes/ 41%
Distribution (Vd): 105 L
Metabolism: cleared predominantly by the CYP3A4 (major route) and CYP2C9 (minor route) hepatic microsomal isoenzymes.
Elimination: 41 L/h

INDICATION:
Treatment of men with erectile dysfunction, which is the inability to achieve or maintain a penile erection sufficient for satisfactory sexual performance.
In order for ELONZA to be effective, sexual stimulation is required.

DOSAGE AND ADMINISTRATION:
Oral
Use in adults
The recommended dose is 50 mg taken as needed approximately one hour before sexual activity. Based on efficacy and toleration, the dose may be increased to 100 mg or decreased to 25 mg. The maximum recommended dosing frequency is once per day. If ELONZA is taken with food, the onset of activity may be delayed compared to the fasted state.
Use in the elderly
Dosage adjustments are not required in elderly patients.
Use in patients with impaired renal function
The dosing recommendations described in “Use in adults” apply to patients with mild to moderate renal impairment (creatinine clearance = 30-80 mL/ min).
Since Sildenafil clearance is reduced in patients with severe renal impairment (creatinine clearance <30 mL/ min) a 25 mg dose should be considered.
Use in patients with impaired hepatic function
Since Sildenafil clearance is reduced in patients with hepatic impairment (e.g. cirrhosis) a 25 mg dose should be considered.
Use in children
ELONZA is not indicated for individuals below 18 years of age. There is no relevant indication for use in children.
Use in patients using other medicines
Given the extent of the interaction with patients receiving concomitant therapy with Ritonavir, it is recommended not to exceed a maximum single dose of 25 mg of Sildenafil in a 48-hour period. A starting dose of 25 mg should be considered in patients receiving concomitant treatment with CYP3A4 inhibitors (e.g. Erythromycin, Saquinavir, Ketoconazole, Itraconazole).

PRECAUTION:
A medical history and physical examination should be undertaken to diagnose erectile dysfunction and determine potential underlying causes, before pharmacological treatment is considered.
Prior to initiating any treatment for erectile dysfunction, physicians should consider the cardiovascular status of their patients, since there is a degree of cardiac risk associated with sexual activity. Sildenafil has a vasodilator property, resulting in mild and transient decreases in blood pressure. Prior to prescribing Sildenafil, physicians should carefully consider whether their patients with certain underlying conditions could be adversely affected by such vasodilatory effects, especially in combination with sexual activity. Patients with increased susceptibility to vasodilators include those with left ventricular outflow obstruction (e.g. aortic stenosis, hypertrophic obstructive cardiomyopathy), or those with the rare syndrome of multiple system atrophy manifesting as severely impaired autonomic control of blood pressure.
ELONZA potentiates the hypotensive effect of nitrates.
Serious cardiovascular events, including myocardial infarction, unstable angina, sudden cardiac death, ventricular arrhythmia, cerebrovascular hemorrhage, transient ischemic attack, hypertension and hypotension have been reported post-marketing in temporal association with the use of ELONZA. Most, but not all, of these patients had pre-existing cardiovascular risk factors. Many events were reported to occur shortly after the use of ELONZA without sexual activity. It is not possible to determine whether these events are related directly to these factors or to other factors.
Agents for the treatment of erectile dysfunction, including Sildenafil, should be used with caution in patients with anatomical deformation of the penis (such as angulations, cavernosal fibrosis or Peyronie’s disease), or in patients who have conditions which may predispose them to priapism (such as sickle cell anemia, multiple myeloma or leukemia).
The safety and efficacy of combinations of Sildenafil with other treatments for erectile dysfunction have not been studied. Therefore the use of such combinations is not recommended.
Visual defects and cases of non-arteritic anterior ischemic optic neuropathy have been reported in connection with the intake of Sildenafil and other PDE5 inhibitors. The patient should be advised that in case of sudden visual defect, he should stop taking ELONZA and consult a physician immediately.
Co-administration of Sildenafil with Ritonavir is not advised.
Caution is advised when Sildenafil is administered to patients taking an alpha-blocker, as the coadministration may lead to symptomatic hypotension in a few susceptible individuals. In order to minimize the potential for developing postural hypotension, patients should be

PM Specification	Product	Code No.	Dimension	Packaging Type	Thickness
	ELONZA 100 mg	IEMY 0042	w. 15.5 x L 25 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm.)
	Designed by: (PDM, ASPD, PDC-M, PDC-A, PDO-M, PDO-A)	Checked by: (PDM/ASPD)		Approved by: (MD (Only Pattern and Colors))	

Approved by: PLC: (Date) LRA: (Date) IRA: (Date) QCC-PM: (Date) CUSTOMER/SALES DEPT.: (Date)

ELONZA 100_Insert_Front,Back_IEMY 0042_By...DA 15/06/2023

BACK 15.5 cm

25 cm

hemodynamically stable on alpha-blocker therapy prior to initiating Sildenafil treatment. Initiation of Sildenafil at a dose of 25 mg should be considered. In addition, physicians should advise patients what to do in the event of postural hypotensive symptoms.
Studies with human platelets indicate that Sildenafil potentiates the antiaggregatory effect of sodium nitroprusside *in vitro*. There is no safety information on the administration of Sildenafil to patients with bleeding disorders or active peptic ulceration. Therefore, Sildenafil should be administered to these patients only after careful benefit-risk assessment.
ELONZA is not indicated for use by women.

PREGNANCY AND LACTATION:
ELONZA is not indicated for use by women.
No relevant adverse effects were found in reproduction studies in rats and rabbits following oral administration of Sildenafil.

SIDE EFFECT:
Very common:
Nervous system disorders (headache)
Vascular disorders (flushing)
Common:
Eye disorders (altered vision and chromatopsia)
Nervous system disorders (dizziness)
Cardiac disorders (palpitations)
Respiratory, thoracic and mediastinal disorders (nasal congestion)
GI disorders (dyspepsia)

CONTRAINDICATION:
Hypersensitivity to the active substance or any of the excipients.
Consistent with its known effects on the nitric oxide/ cyclic guanosine monophosphate (cGMP) pathway, Sildenafil was shown to potentiate the hypotensive effects of nitrates, and its co-administration with nitric oxide donors (such as amyl nitrite) or nitrates in any form is therefore contraindicated.

DRUG INTERACTION:
Effects of other medicinal products on Sildenafil
Sildenafil metabolism is principally mediated by the cytochrome P450 (CYP) isoforms 3A4 (major route) and 2C9 (minor route). Therefore, inhibitors of these isoenzymes may reduce Sildenafil clearance and inducers of these isoenzymes, may increase Sildenafil clearance. Efficacy of Sildenafil should be closely monitored in patients using concomitant potent CYP3A4 inducers, such as Carbamazepine, Phenytoin, Phenobarbital, St. John's wort and Rifampicin. This is consistent with Ritonavir's marked effects on a broad range of P450 substrates. Based on these pharmacokinetic results co-administration of Sildenafil with Ritonavir is contraindicated in pulmonary arterial hypertension patients. Sildenafil had no effect on Saquinavir pharmacokinetics. Potent CYP3A4 inhibitors such as Ketoconazole and Itraconazole would be expected to have effects similar to Ritonavir. CYP3A4 inhibitors of intermediate potency (e.g. Clarithromycin, Telithromycin and Nefazodone) are expected to have an effect in between that of Ritonavir and CYP3A4 inhibitors of medium potency (e.g. Saquinavir/ Erythromycin), a seven-fold increase in exposure is assumed. Therefore dose adjustments are recommended when using CYP3A4 inhibitors of intermediate potency. The population pharmacokinetic analysis in pulmonary arterial hypertension patients suggested that co-administration of beta-blockers in combination with CYP3A4 substrates might result in an additional increase in Sildenafil exposure compared with administration of CYP3A4 substrates alone. Grapefruit juice is a weak inhibitor of CYP3A4 gut wall metabolism and may give rise to modest increases in plasma levels of Sildenafil. Single doses of antacid (Magnesium hydroxide/ Aluminum hydroxide) did not affect the bioavailability of Sildenafil. Co-administration of oral contraceptives (Ethinyloestradiol 30 mg and Levonorgestrel 150 mg) did not affect the pharmacokinetics of Sildenafil. Nicorandil is a hybrid of potassium channel activator and nitrate. Due to the nitrate component, it has the potential to have serious interaction with Sildenafil. No significant interactions were shown when Sildenafil (50 mg) was co-administered with Tolbutamide (250 mg) or Warfarin (40 mg), both of which are metabolized by CYP2C9. Sildenafil had no significant effect on Atorvastatin exposure (AUC increased 11%, suggesting that Sildenafil does not have a clinically relevant effect on CYP3A4. No interactions were observed between Sildenafil (100 mg single dose) and Acenocoumarol. Sildenafil (100 mg single dose) did not affect the steady state pharmacokinetics of the HIV protease inhibitor Saquinavir, which is a CYP3A4 substrate/ inhibitor. Consistent with its known effects on the nitric oxide/ cGMP pathway, Sildenafil was shown to potentiate the hypotensive effects of nitrates in any form is therefore contraindicated. Sildenafil had no clinically significant impact on the plasma levels of oral contraceptives (Ethinyloestradiol 30 mg and Levonorgestrel 150 mg).

OVERDOSE AND TREATMENT:
In single dose volunteer studies of doses up to 800 mg, adverse reaction were similar to those seen at lower doses, but the incidence rates and severities were increased. Doses of 200 mg did not result in increased efficacy but the incidence of adverse reactions (headache, flushing, dizziness, dyspepsia, nasal congestion, altered vision) was increased.
In cases of overdose, standard supportive measures should be adopted as required. Renal dialysis is not expected to accelerate clearance as Sildenafil is highly bound to plasma proteins and not eliminated in the urine.

STORAGE:
Store at temperature of not more than 30°C.

DOSAGE FORM AND PACKAGING AVAILABLE:
100 mg Tablet, Blister 10x2's; 2x2's

DATE OF REVISION:
June 13, 2023

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

C-MY/130623-07 (AR)
(www.emc.medicines.org.uk)

IEMY 0042

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